

1(a)	acceleration: vector electrical resistance: scalar momentum: vector <i>1 mark for two correct, 2 marks for all three correct</i>	B2
1(b)	resultant force (in any direction) is zero	B1
	resultant torque/moment (about any point) is zero	B1
1(c)(i)	upthrust = $\rho g(\Delta)h \times A$	C1
	$= (1.00 \times 10^3 \times 9.81 \times 0.190) \times 0.0230$ $= 42.9 \text{ N}$	A1
1(c)(ii)	($T =$) $43 - 28 = 15 \text{ (N)}$ or ($T =$) $42.9 - 28 = 14.9 \text{ or } 15 \text{ (N)}$	A1
1(c)(iii)	$\sigma = F / A$ or T / A	C1
	$= 15 / (3.2 \times 10^{-6})$ $= 4.7 \times 10^6 \text{ Pa}$	A1
1(c)(iv)	upthrust (on cylinder) increases (and weight constant)	B1
	tension/stress increases and (so) strain energy increases	B1